

Course: MCA

Semester: 5

Prerequisite: Fundamental knowledge of Object oriented programming, DBMS and HTML.

Course Objective: To provide basic understanding of .NET framework and knowledge of developing dynamic and rich web application in conjunction with event handling, state management and data access.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
4	0	4	-	6	20	20	20	60	30	150

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	Introduction ASP.NET framework, The origin of .NET technology, Common Language Runtime (CLR), Managed / Unmanaged code, Common Type System (CTS), Common Language Specification (CLS), Microsoft Intermediate Language (MSIL), Just-in-time compiler, Framework base classes, Assemblies – Private and Shared, Garbage Collection	10	7
2	ASP.NET Controls, UI Design & Coding ASP.NET server controls: Controls - buttons, textboxes, labels, checkboxes, radio buttons, list controls and other web server controls, ImageMap Control, web.config and global.asax files. Programming ASP.NET web pages: Web page, Data types and variables, Statements, Organizing code, Object oriented basics of C#.	25	13
3	ASP.NET Validation Controls, State Management & Navigation Validation Control: Controls, Basic validation controls, Validation techniques, Using advanced validation controls. State Management: Using view state, Using session state, Using application state, Using cookies and URL encoding. Master Pages: Creating master pages, Content pages, nesting master pages, Accessing master page controls from a content page. Themes : Skin File and CSS Navigation: Site navigation, Using site navigation controls.	25	16
4	Data Access through ADO.NET Basic features of ADO.NET, ADO.NET Architecture, Using SQL data sources, Grid view control, Detail view control, Form view control, List view and other controls, Using object data sources.	20	11
5	Advanced ASP.NET AJAX extension, LINQ, Stored procedure, Working with XML data, Captcha Control. ASP.NET MVC: Overview of Model-View-Controller, Model-View-Controller and ASP.NET, Routes and URLs, Controllers, Views.	20	13

Reference Books

1.	Professional ASP.Net By Bill Evjen, Scott Hanselman, Devin Rader Wrox Publication
2.	ASP.NET 4 Unleashed By Stephen Walther, Kevin Hoffman, Nate Dudek Sams Publication
3.	NET 4.0 Programming Black Book Kogent Learning Solutions Incorporation

Course Outcome

After Learning the Course the students shall be able to:

1. Explain .NET framework architecture and its features.
2. Build dynamic web applications to demonstrate event handling, state management and content manipulation using C#.
3. Develop data driven applications using ADO.NET, LINQ and XML.
4. Construct web applications using model-view-controller pattern.

List of Practical

1.	Label and Button controls Create a web application to display "Welcome to the world of ASP.NET" on web browser when user "Click" button is clicked.
2.	Student Marksheet using web controls Create a web page in which user can enter enrollment no, student name and marks of 5 subjects. Calculate total, percentage and grade of student and display it. [Use text box, label, button control].
3.	Textbox and Button controls Write a program to enable - disable textbox and change width of text box programmatically.
4.	Use of Radio button Create a web page that will take two radio buttons labeled as a gender, if user selects radio button the gender will be changed according to selected radio button.
5.	List box and Image controls Write a program which have list box and image controls, list box is used to display items and when user clicks on item in list box it display the item image in image control.
6.	Registration page Create a web page for registration which have name, dob, gender, email id, contact no, address, city, pin code, hobbies and display all data in same page with proper layout.
7.	Required Field validation Write a program that uses a textbox for a user input and validate it using required field validation.
8.	Registration page with validation Create a web page for registration which have name, dob, gender, email id, contact no, address, city, pin code, hobbies. After the user enters the appropriate values the validate button must validates the values entered and display all data in same page with proper layout.
9.	Custom Validation Write a program to check the length of the string in the text box using custom validation.
10.	Introduction to Master page Create a master page that contain links for home page, about us page and contact page, when user clicks on particular link it will display that particular page.
11.	Use of Master page and Session Create a web application for registration system that contains navigation control to specify link like registration and login. When user click registration link user should be navigated to registration page. When user click on the login link login page will be display that will redirected to registration page if user logged in to system. [Use master page and session to maintain

	the user identity].												
12.	Cookie's Write a program that has a form taking the user's name as input. Store this name in a permanent cookie & whenever the page is opened again, then value of the name field should be attached with the cookie's content.												
13.	Database CRUD operations [Gridview] Create student database which contain table stud_registration and have fields like first name, last name, dob, email id, contact number, gender, address, city, pin code. Create web application which will insert, view, search, update, and delete records from stud_registration table. [Use gridview]												
14.	Database CRUD operations [Listview] Create product data base which contain product_info table and have fields like product id, product name, product quantity, price . Create a web application with validation which will insert, view, search, update and delete records from table. [Use listview]												
15.	Database Operations Create table employee with the following columns and data types. <table border="1"> <thead> <tr> <th>Columns</th><th>Data Type</th></tr> </thead> <tbody> <tr> <td>dept_id</td><td>int</td></tr> <tr> <td>Name</td><td>nvarchar(20)</td></tr> <tr> <td>DOJ</td><td>datetime</td></tr> <tr> <td>Salary</td><td>float</td></tr> <tr> <td>Designation</td><td>nvarchar(30)</td></tr> </tbody> </table> - Insert following records into the table - Change the candidate name from "Ansh" to "Alpesh". - Select records from table where salary > 10000 and display it.	Columns	Data Type	dept_id	int	Name	nvarchar(20)	DOJ	datetime	Salary	float	Designation	nvarchar(30)
Columns	Data Type												
dept_id	int												
Name	nvarchar(20)												
DOJ	datetime												
Salary	float												
Designation	nvarchar(30)												
16.	LINQ and Stored Procedure Create database train_master having table train_info and have fields like train_id (PK, AI), train_name, train_type, arrival_time, departure_time, start_location, end_location. Create web page traindata.aspx that allow user to enter train_id, train_name, train_type, arrival_time, departure_time, start_location, end_location and perform following operations Insert, delete, view, update, search [Perform this application using LINQ and Stored Procedure]												
17.	Ajax Create a Calendar Navigation by using Ajax Calendar Extender control ,in which user can Navigate Year , Month and select date from the Calendar control.												

18.	XML Create an application which reads data from an XML file and also write data to the file.
19.	Introduction to MVC Create database book_master having table book_info and have fields like book_id (PK, AI), book_name, author, price. Create a web application with validation which will insert, view, search, update and delete records from table. [Use MVC architecture]
20.	Mini Project Create a data driven website system from following list. • College Web Portal • E shopping • Online Student Management • Online Inventory • Web Time Sheet • Complain Management • Alumni Information System • Property Management • Online Hotel Booking • Library Management • Hospital Management